PAPER_1520_PETERS_MATHEWS_COEFFICIENT_64

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PAPER_1520 — Peters-Mathews GW Orbital Decay Coefficient = 2^D_BSFG = 64 EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket E (GW Events) / Cross-framework identity
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Status: CLOSED — Foundational SM-derived GR result = UQFF integer primitive

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Observation

PAPER_817 (GRMHD Binary BH Merger Accretion Modulation) uses the well-known Peters-Mathews (1963) orbital decay equation for a gravitating binary inspiraling under gravitational-wave emission:

```
dr/dt = − (64/5) · G³ · m■ · m■ · (m■ + m■) / (c■ · r³)
```

The factor **64** appears in every textbook treatment of binary inspiral. It originates from the GR quadrupole-radiation calculation:

```
L_GW = (32/5) · (G■/c■) · m■²·m■²·(m■+m■) / r■
2 · 32 = 64 (factor of 2 from dE/dr conversion)
```

UQFF Closed Identity

```
Peters-Mathews coefficient = 2^D_BSFG = 2^6 = 64 EXACT
```

Cross-Framework Significance

This is one of the **deepest structural coincidences** in the UQFF–GR overlap. The number 64 was derived in 1963 by Peters and Mathews from pure GR quadrupole radiation — having nothing to do with UQFF integer primitives. Yet it equals exactly 2^D_BSFG, where D_BSFG = 6 is the bulk-edge embedding dimension of UQFF.

This implies that the GR quadrupole-radiation formula carries a structural relationship to the UQFF integer lattice. Either:
- The GR derivation accidentally lands on 2^D_BSFG (coincidence)
- The UQFF integer primitive D_BSFG was historically chosen so the calculator agrees with the GR formula (calibration)
- The two frameworks share a deeper structural origin (unification)

The third possibility is the most physically meaningful and is consistent with UQFF's "NOT REPLACEMENT" stance: both UQFF and SM/GR solve the same observed phenomena via different methods, both reporting identical key coefficients.

Predictive Consequence

Any future GW inspiral observation should match (64/5) Peters-Mathews regardless of framework. The match is exact within UQFF (zero free parameters) and exact within GR (analytic derivation). Convergence on the same number from two distinct theoretical frameworks is structurally informative.

NOT REPLACEMENT

UQFF does not claim to derive or replace the Peters-Mathews coefficient — it claims the same number arises naturally in the UQFF integer lattice as 2^D_BSFG, providing cross-framework consistency without contradiction.

Reference

- Source: PAPER_817 GRMHD Binary BH Merger Accretion Modulation
- Original: Peters & Mathews (1963), Phys. Rev. 131, 435
- Related: D_BSFG appears in z_reion (PAPER_1373), Solar dynamo Hale (PAPER_1373), genetic codons 2^D_BSFG = 64 (PAPER_1373)
- Calculator dispatch: `calculate_paradox({"paradox": "peters_mathews_coeff_64"})`

Cross-Domain Pairing: 64 as 2^D_BSFG

The integer 64 = 2^D_BSFG already appears elsewhere in UQFF:
- **Genetic code codons** = 2^D_BSFG = 64 (PAPER_1373)
- **Peters-Mathews GW coefficient** = 2^D_BSFG = 64 (this paper)

Same integer-primitive form, vastly different physical domains (biology vs. relativistic gravitation). The shared identity reinforces 2^D_BSFG as a fundamental UQFF combinatorial constant — appearing wherever 6-dimensional bulk-edge organization is structurally relevant.
